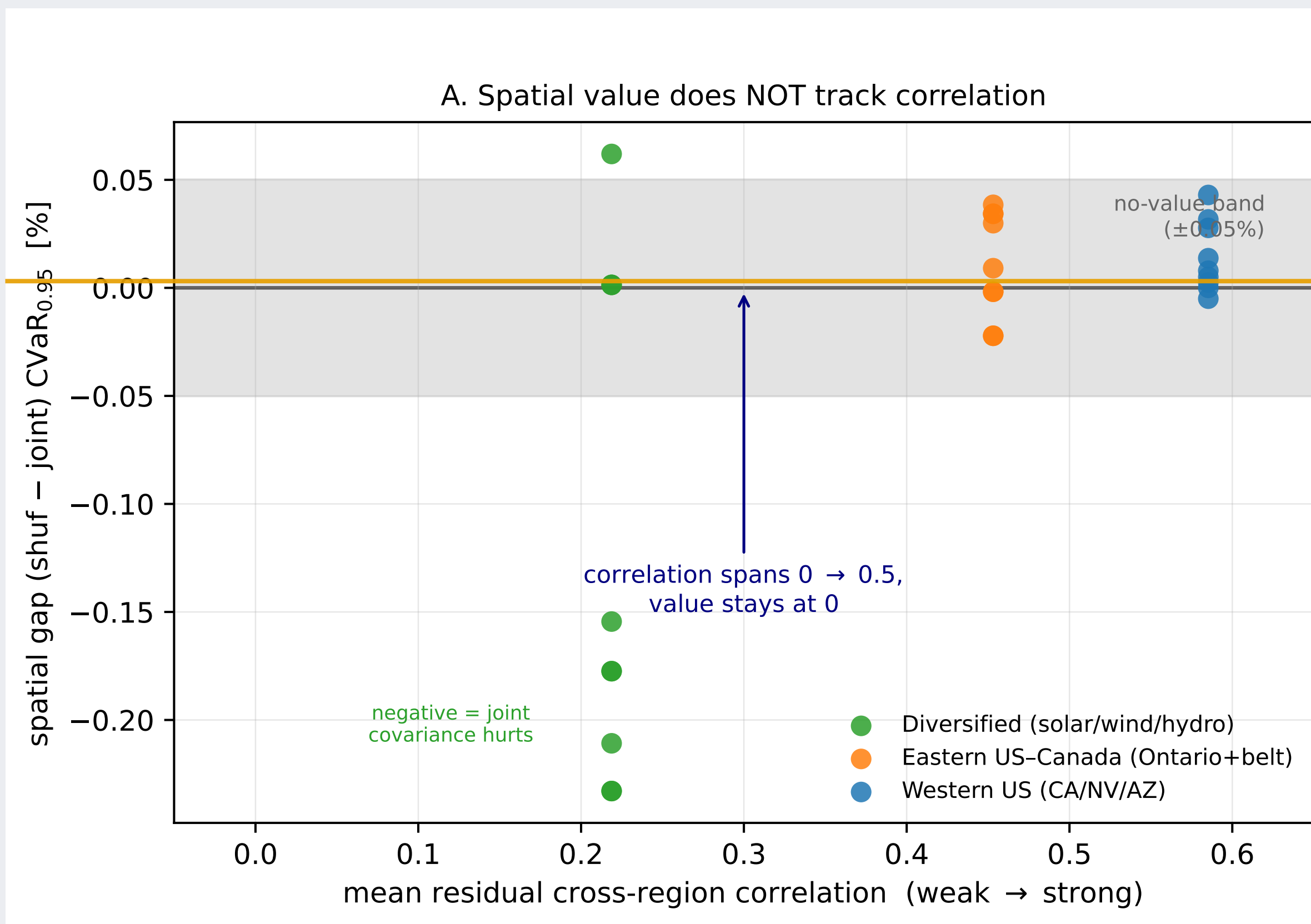




Does Spatial Correlation of Carbon Intensity Improve Robust Carbon-Aware Data-Center Scheduling?

A Multi-Grid Falsification Study: Causal Mechanism, Copula Test, and an Active-Transfer Extension

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Across every grid and every model, the gap never leaves **zero**.

The diurnal **mean** carbon field, not its spatial **shape**, is what a robust scheduler can act on.

0.78

CROSS-GRID CORRELATION

IS REAL

~0%

ROBUST SCHEDULING

VALUE

+1.46%

ONLY WHEN THE

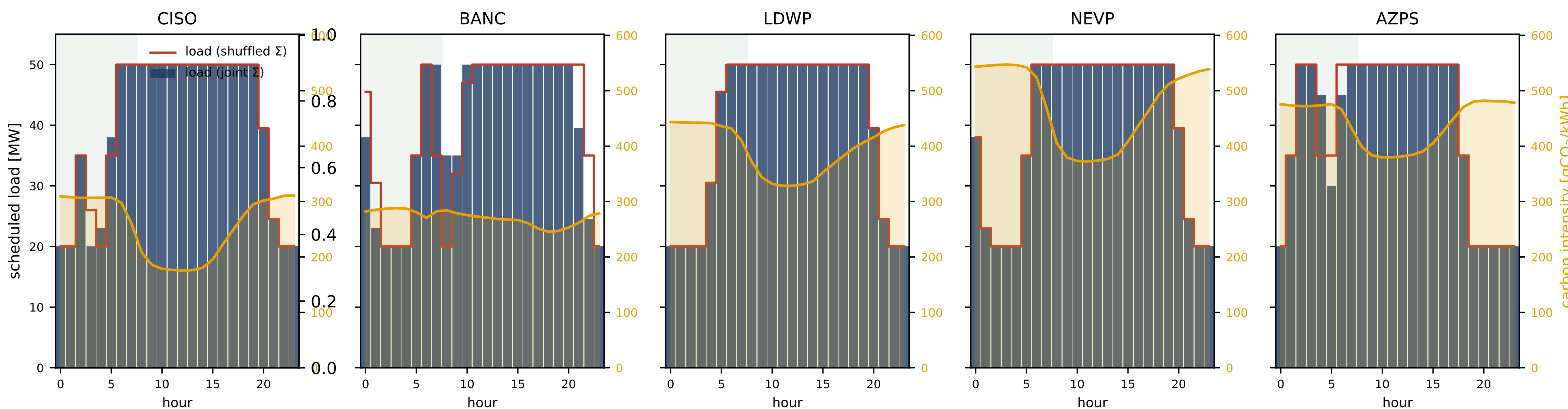
MEAN IS FLATTENED

4-10%

ONLY VIA

ACTIVE TRANSFER

Joint vs. shuffled-covariance schedules are visually indistinguishable (Western US): the spatial null, seen directly. Load (bars) tracks the low-carbon hours.



The setup

- Data centers consume 1–2% of global electricity, and compute is unusually **flexible**: batch/training jobs can shift in time and across sites.
- Carbon intensity varies strongly by hour and region, so a scheduler can defer work to cleaner moments.
- Robust scheduling under *uncertain* carbon intensity uses **distributionally robust optimization (DRO)**.
- A natural extension treats carbon intensity as a stochastic *vector* so that **spatial correlation** can inform the schedule.

RQ1. Is carbon intensity of coupled grids genuinely correlated?

RQ2. Does feeding that correlation to a covariance-based DRO reduce out-of-sample tail emissions (CVaR_{0.95})?

RQ3. If not, *why*, and what dependence structure would a richer model need?

Value is

The spatial gap is not merely s across the whole dependence s

- Survives Ledoit–Wolf shrinkage, and the full tail
- Survives Benjamini–Hochberg and walk-forward testing to
- A per-cell **equivalence test** rejects any effect above the **0** 27 cells.
- The gap is statistically = **0**.

This is an **active null**: CV se cell, so the DRO is genuinely variance *still* buys nothing.

The r

- **Mean-ablation** and the joint cov The spatial signa mean field, not a
- **Tail dependence** is **non-elli** than *dirty* togeth An elliptical cov *wrong object*.

How we test it

A Mahalanobis–Wasserstein DRO, solved as a second-order cone program:

$$\min_{x \in X, t \geq 0} \langle \bar{\rho}, x \rangle + \varepsilon t \quad \text{s.t.} \quad \|L^\top \text{vec}(x)\|_2 \leq t, \quad LL^\top = \hat{\Sigma}.$$

Shuffled-marginals test.

- Fit the schedule twice, on the *joint* covariance vs. a *block-diagonal* one with all cross-region structure zeroed, and compare out-of-sample CVaR_{0.95} of daily emissions.
- The *spatial gap* is positive iff the joint covariance helps.
- Pre-registered: config commit-locked, single 2025 test read

